Tora: Trajectory-oriented Diffusion Transformer for Video Generation

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Figure 1. Tora is capable of generating videos guided by arbitrary trajectories, images, texts, or combinations thereof. Our proposed motion modules integrate seamlessly with the scalability of DiT, ensuring that the generated movements not only adhere precisely to the specified trajectory but also effectively emulate physical world dynamics.

Abstract

Recent advancements in Diffusion Transformer (DiT) have demonstrated remarkable proficiency in producing high-quality video content. Nonetheless, the potential of transformer-based diffusion models for effectively generating videos with controllable motion remains an area of limited exploration. This paper introduces Tora, the first trajectory-oriented DiT framework that concurrently integrates textual, visual, and trajectory conditions, thereby enabling scalable video generation with effective motion guidance. Specifically, Tora consists of a Trajectory Extractor (TE), a Spatial-Temporal DiT, and a Motion-guidance Fuser (MGF). The TE encodes arbitrary trajectories into hierarchical spacetime motion patches with a 3D motion compression network. The MGF integrates the motion patches into the DiT blocks to generate consistent videos that accurately follow designated trajectories. Our design aligns seamlessly with DiT's scalability, allowing precise control of video content's dynamics with diverse durations, aspect ratios, and resolutions. Extensive experiments demonstrate that Tora excels in achieving high motion fidelity compared to the foundational DiT model, while also accurately simulating the complex movements of the physical world. Code is made available at https://github.com/alibaba/Tora .

1. Introduction

Diffusion models [9, 29] have demonstrated their capability to generate diverse and high-quality images or videos. Previously, video diffusion models [3, 13, 47] predominantly employed UNet-based architectures [31], focusing primarily on synthesizing videos of limited duration, typically around two seconds, and were constrained to fixed resolutions and aspect ratios. Recently, Sora [4], a text-to-video generation model leveraging Diffusion Transformer (DiT) [25], has showcased video generation capabilities that significantly outstrip current state-of-the-art methods. Sora excels not only in the production of high-quality videos ranging from 10 to 60 seconds, but also distinguishes

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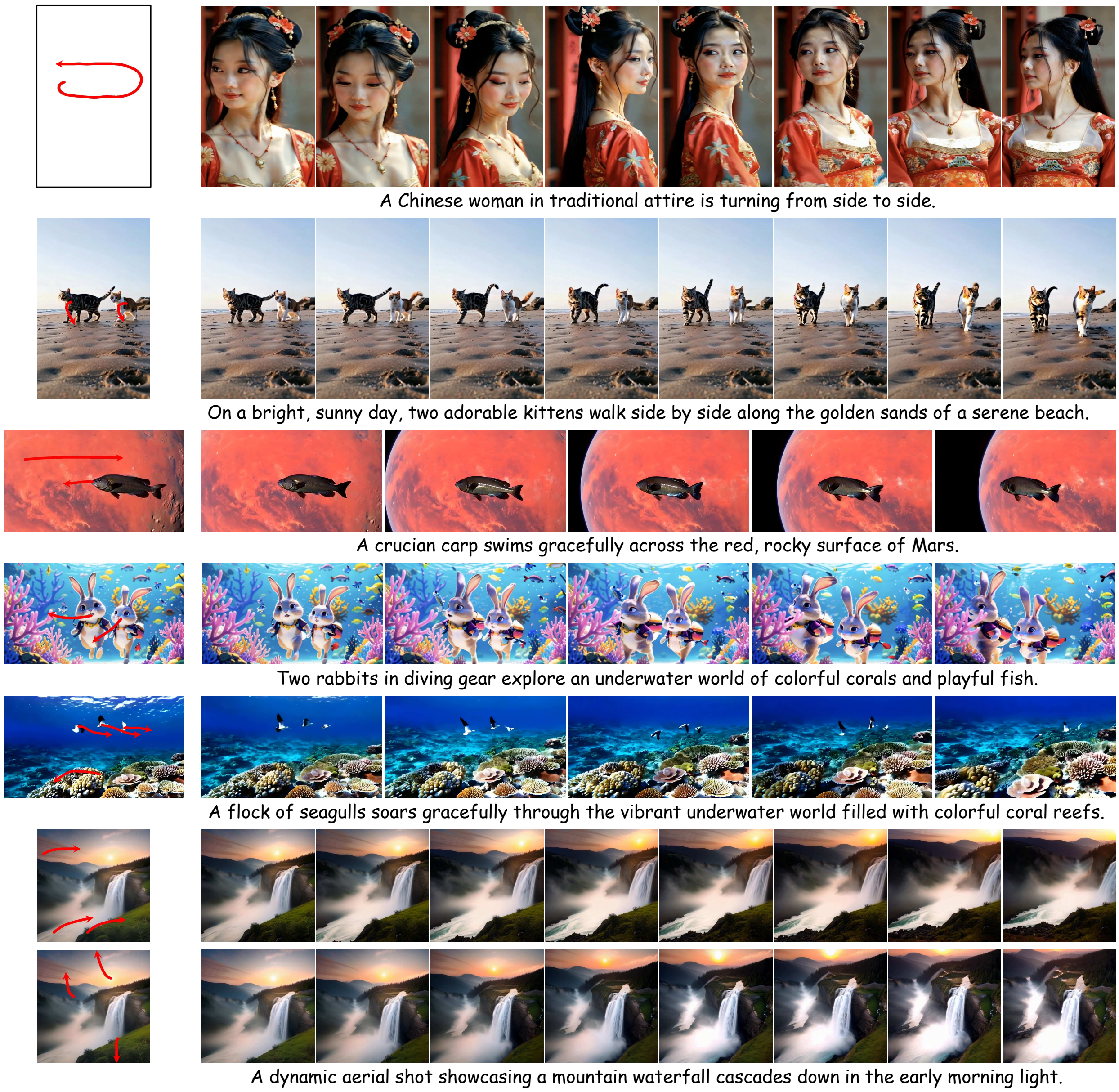

Figure 2. More generated samples. Tora effectively manages trajectories to precisely manipulate various objects and backgrounds. In the realm of image-to-video synthesis, it can craft dynamic camera movements in accordance with textual descriptions and the designated starting trajectory points, such as common backgrounds. Furthermore, Tora supports video generation across different aspect ratios, resolutions, and durations, ensuring flexible content creation.

itself through its capacity to handle diverse resolutions, various aspect ratios, adherence to the laws of actual physics.

Video generation requires consistent motion across image sequences, underscoring the importance of motion control. Previous works, such as VideoComposer [36] and DragNUWA [44], have implemented generalized motion manipulation through motion vectors and trajectories. Building on this foundation, MotionCtrl [38] innovates by independently managing camera and object motions, thereby expanding the diversity of achievable motion patterns. Despite their promising controllable motion quality, UNet-based methods are restricted to generating videos of only 16 frames at a fixed, lower resolution. This limitation hinders the smooth portrayal of motion, particularly during significant positional shifts in the provided trajectory, leading to distortion and unnatural movements, such as parallel drifting, which diverge from real-world dynamics. Consequently, there is an urgent need for a model capable of producing longer videos with robust motion control and detailed physical representations.

To address these challenges, we present Tora, the first DiT model that simultaneously integrates text, images, and trajectories, enabling scalable video generation with robust motion control. Notably, our work adopts OpenSora [51], an open-source version of Sora, as the foundational DiT model. To align motion control with the scalability of the DiT framework, we propose two novel modules: the Trajectory Extractor (TE), which converts arbitrary trajectories into hierarchical spacetime motion patches, and the Motion-guidance Fuser (MGF), designed to seamlessly integrate these patches within the stacked DiT blocks. More specifically, TE initially converts positional displacements along trajectory into the RGB domain via flow visualization techniques. These visualized displacements undergo Gaussian filtering to mitigate scattered issues. Subsequently, a 3D Variational Autoencoder (VAE) [18] encodes trajectory visualizations into spacetime motion latents, which share the same latent space with video patches. The motion latents are then decomposed into multiple levels of motion conditions via stacked lightweight modules. Our VAE architecture is inspired by MAGVIT-v2 [46] but simplified by omitting codebook dependencies. The MGF integrates adaptive normalization layers [26] to infuse multi-level motion conditions into the corresponding DiT blocks. We explored various adaptations of transformer blocks including adaptive layer normalization, cross-attention, and extra channel connections to inject the motion conditions. Among these, adaptive layer normalization emerged as the most effective to generate consistent videos following the trajectory.

During training, we adapt OpenSora's workflow to generate high-quality video-text pairs and utilize an optical flow estimator [41] for trajectory extraction. We also integrate a motion segmentor [50] with a camera detector[1] to filter out instances dominated by camera motion, thereby improving our tracking of object trajectories. This careful selection process results in a dataset of high-quality videos with consistent motion. With an adapter-like strategy [23], we solely train the temporal blocks, together with the TE and MGF. This strategy seamlessly integrates DiT's inherent generative knowledge with external motion signals.

The main contributions of our work are as follows:

- We introduce Tora, the first trajectory-oriented DiT model for video generation. As illustrated in Figure 2, Tora enables the creation of motion-manipulable videos with varying aspect ratios, extending up to 204 frames and 720p resolution.
- We propose a novel trajectory extractor and a motion-guidance fusion mechanism to facilitate motion control that aligns with the scalability of DiT. Additionally, we ablate several architectural choices and scaling capabilities to offer empirical baselines for future research.
- Experiments demonstrate that Tora achieves state-of-the-art accuracy in controlling object motions. Furthermore, it demonstrates superiority in simulating movements within the physical world.

[1]https://github.com/antiboredom/camera-motion-detector

2. Related Work

2.1. Diffusion models for Video Generation

Diffusion models have demonstrated an impressive capability to generate high-quality video samples. Previous research [12, 13, 16, 33, 48] commonly use video diffusion models (VDMs) that incorporate temporal convolutional and attention layers into the pre-trained image diffusion models. Subsequently, VideoCrafter [6] and SVD [3] expand the application of video diffusion models to larger datasets, while TF-T2V [37] directly learns from extensive text-free videos. Nonetheless, these methods encounter limitations in generating long videos, owing to the inherent constraints on capacity and scalability within the UNet design. Conversely, DiT-based models [2, 4, 28, 43, 51] can directly generate videos extending up to tens of seconds. Sora [4] converts visual data into a unified representation, facilitating large-scale training and enabling the generation of 1-minute high-definition video. Vidu [2] is capable of generating both realistic and imaginative videos in various aspect ratios. CogVideoX [43] introduces an expert diffusion transformer model that generates videos from text prompts or images, along with an effective text-video data processing pipeline that enhances video caption quality.

2.2. Motion control in Video Generation

To better control motion in generated videos, a multitude of studies have endeavored to introduce diverse motion signals in VDMs. Pioneering works like MotionDirector [49] and VMC [15] have utilized reference videos to extract motion patterns applicable to diverse video generations. VideoComposer [36] expands upon this by adopting depth maps, sketches, or motion vectors from references as conditional inputs for motion control. Nonetheless, these methodologies are limited to reproducing existing motion patterns. Conversely, approaches that leverage trajectories [11, 19, 22, 32, 38, 44] or bounding boxes [8, 38, 44] in video generation promise greater adaptability and user accessibility. DragNUWA [44] breaks new ground by integrating trajectory-based conditioning into VDMs, facilitating complex camera and object movements. AnimateAnything [8] employs motion masks for precise control over the moving regions. DragAnything [40] uses the object mask to generate entity representations for achieving motion control. MotionCtrl [38] facilitates more flexible control, allowing separate adjustment of both camera movements and

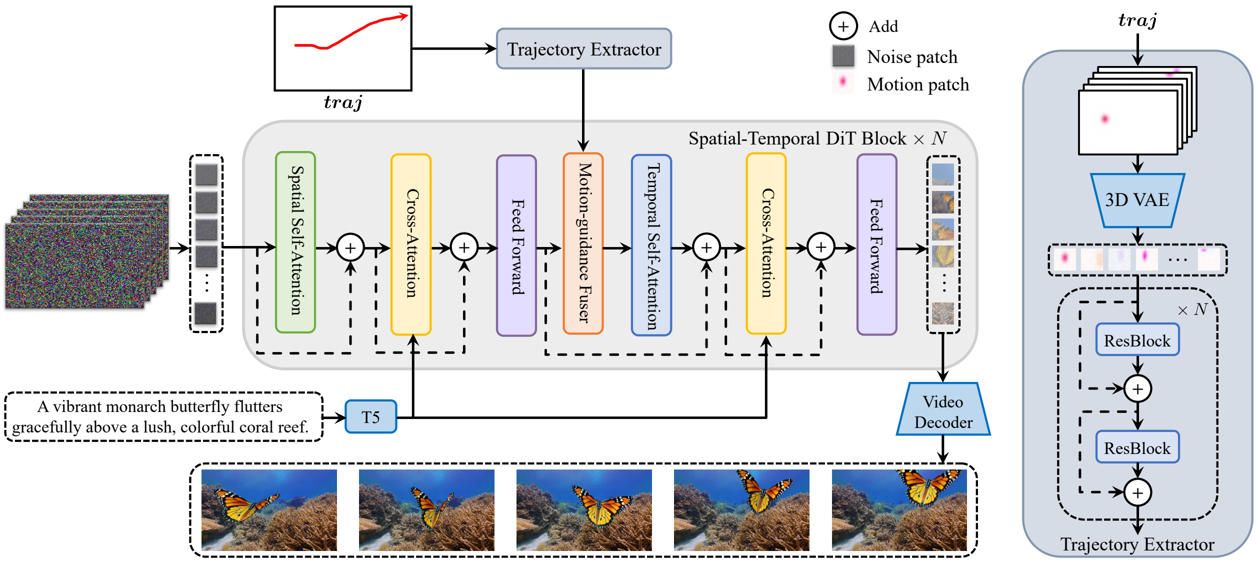

Figure 3. Overview of the Tora Architecture. We introduce two novel modules: the Trajectory Extractor and the Motion-guidance Fuser. The Trajectory Extractor uses a 3D motion VAE to embed trajectory vectors into the same latent space as video patches, preserving motion information across frames. It then employs stacked convolutional layers to extract hierarchical motion features. The Motion-guidance Fuser utilizes adaptive normalization layers to integrate these multi-level motion conditions into the corresponding DiT blocks, ensuring that generated videos consistently follow defined trajectories. Our method leverages the scalability of DiT, enabling the creation of motion-controllable videos of extended duration.

individual object motions. However, all of them yield noticeable artifacts in both motion consistency and visual presentation when applied to longer sequences. In contrast, our method first integrates trajectories into DiT architecture, which enables closer adherence to the physical world.

3. Methodology

3.1. Preliminary

Latent Video Diffusion Model (LVDM). The LVDM enhances the stable diffusion model [29] by integrating a 3D UNet, thereby empowering efficient video data processing. This 3D UNet design augments each spatial convolution with an additional temporal convolution and follows each spatial attention block with a corresponding temporal attention block. It is optimized by employing a noise-prediction objective function:

$$l_\epsilon = ||\epsilon - \epsilon_\theta(z_t, t, c)||_2^2, \tag{1}$$

Here, $\epsilon_\theta(\cdot)$ signifies the 3D UNet's noise prediction function. The condition c is guided into the UNet using cross-attention for adjustment. Meanwhile, z_t denotes the noisy hidden state, evolving like a Markov chain that progressively adds Gaussian noise to the initial latent state z_0:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \tag{2}$$

where $\bar{\alpha}_t = \prod_{i=1}^{t}(1 - \beta_t)$ and β_t is a coefficient that controls the noise strength in step t.

Diffusion Transformer (DiT). The DiT [25] introduces a novel architecture that merges the strengths of diffusion models with transformer architectures [35]. This integration aims to address the limitations of traditional UNet-based latent diffusion models (LDMs), improving their performance, versatility, and scalability. While keeping the overall framework consistent with existing LDMs, the key shift lies in replacing the UNet with a transformer architecture for learning the denoising function $\epsilon_\theta(\cdot)$, thereby marking a pivotal advance in the realm of generative modeling.

3.2. Tora

Tora employs the Spatial-Temporal Diffusion Transformer (ST-DiT) from OpenSora as its foundational model. To facilitate user-friendly motion control while aligning with the scalability of DiT, Tora integrates two novel motion-processing components: the Trajectory Extractor (TE) and the Motion-guidance Fuser (MGF). An overview of Tora's workflow is illustrated in Figure 3.

Spatial-Temporal DiT. The ST-DiT architecture incorporates two distinct block types: the Spatial DiT Block (S-DiT-B) and the Temporal DiT Block (T-DiT-B), arranged in an alternating sequence. The S-DiT-B comprises two attention layers, each performing Spatial Self-Attention (SSA) and Cross-Attention sequentially, succeeded by a point-wise feed-forward layer that serves to connect adjacent T-DiT-B block. Notably, the T-DiT-B modifies this schema solely by substituting SSA with Temporal Self-Attention

(TSA), preserving architectural coherence. Within each block, the input, upon undergoing normalization, is concatenated back to the block's output via skip-connections. By leveraging the ability to process variable-length sequences, the denoising ST-DiT can handle videos of variable durations.

During processing, a video autoencoder [45] is first employed to diminish both spatial and temporal dimensions of videos. To elaborate, it encodes the input video $X \in \mathbb{R}^{L\times H\times W\times 3}$ into video latent $z_0 \in \mathbb{R}^{l\times h\times w\times 4}$, where L denotes the video length and $l = L/4, h = H/8, w = W/8$. z_0 is next "patchified", resulting in a sequence of input tokens $I \in \mathbb{R}^{l\times s\times d}$. Here, $s = hw/p^2$ and p denotes the patch size. In both SSA and TSA, standard Attention is performed using Query (Q), Key (K), and Value (V) matrices:

$$Q = W_Q \cdot I_{\text{norm}}; K = W_K \cdot I_{\text{norm}}; V = W_V \cdot I_{\text{norm}}, \quad (3)$$

Here, I_{norm} is the normalized I, W_Q, W_K, W_V are learnable matrices. The textual prompt is embedded with a T5 encoder and integrated using a cross-attention mechanism.

Trajectory Extractor. Trajectories have proven to be a more user-friendly method for controlling the motion of generated videos. Specifically, given a trajectory $traj = \{(x_i, y_i)\}_{i=0}^{L-1}$, where (x_i, y_i) denotes the spatial position (x, y) at the $i\text{-}th$ frame the trajectory passes through. Previous studies primarily encode the horizontal offset $u(x_i, y_i)$ and the vertical offset $v(x_i, y_i)$ as the motion condition:

$$u(x_i, y_i) = x_{i+1} - x_i;\; v(x_i, y_i) = y_{i+1} - y_i, \quad (4)$$

However, the DiT model employs a video autoencoder and a patchification process to convert the video into patches. Here, each patch is derived across multiple frames, rendering it inappropriate to straightforwardly employ frame-to-frame offsets. To address this, our TE converts the trajectory into motion patches, which inhabit the same latent space as the video patches. Particularly, we begin by transforming the $traj$ into a trajectory map $g \in \mathbb{R}^{L\times H\times W\times 2}$, enhanced with a Gaussian Filter to mitigate scatter. Notably, the first frame employs a fully-zero map. Afterward, the trajectory map g is translated into the RGB color space, producing $g_{vis} \in \mathbb{R}^{L\times H\times W\times 3}$ through a flow visualization technique. We use a 3D VAE to compress trajectory maps, achieving an 8x spatial and 4x temporal reduction, aligning with Open-Sora framework. Our VAE is based on the MAGVIT-v2 architecture, with spatial compression initialized using the VAE of SDXL [27] to accelerate convergence. We train the model using reconstruction loss to obtain the compact motion latent representation $g_m \in \mathbb{R}^{l\times h\times w\times 4}$ from the g_{vis}.

To match the size of the video patches, we use the same patch size on g_m and encode it using a series of convolutional layers, resulting in spacetime motion patches $f \in \mathbb{R}^{l\times s\times d'}$. Here d' is the dimension of motion patches. The

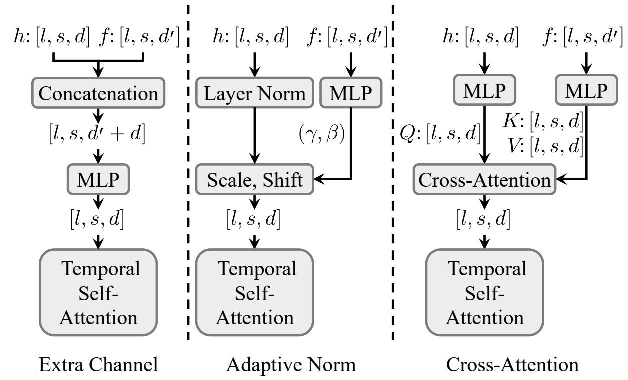

Figure 4. Different designs of the Motion-guidance Fuser for incorporating trajectory conditioning. Adaptive Norm demonstrates the best performance.

output of each convolutional layer is skip-connected to the input of the next layer to extract multi-level motion features:

$$f_i = \text{Conv}^i\,(f_{i-1}) + f_{i-1}, \quad (5)$$

where f_i is the motion condition for $i\text{-}th$ ST-DiT block.

Motion-guidance Fuser. To incorporate DiT-based video generation with the trajectory, we explore three variants of fusion architectures that inject motion patches into each ST-DiT block. These designs are illustrated in Figure 4.

- Extra channel connections. Denote $h_i \in \mathbb{R}^{l\times s\times d}$ as the resultant output from the $i\text{-}th$ block of the ST-DiT. Following the widespread use of concatenation in GAN-based LVDM, the motion patches are simply concatenated with the previous hidden state h_{i-1} along the channel dimension. An additional MLP is then added to maintain the same latent size:

$$h_i = \text{MLP}\,([h_{i-1}, f_i]) + h_{i-1}, \quad (6)$$

- Adaptive Norm layer. Inspired by the adaptive normalization layers employed in GANs, we initially convert f_i into scale γ_i and shift β_i by adding two zero-initialized convolution layers into the ST-DiT block. Subsequently, γ_i and β_i are integrated into h_i through a straightforward linear projection:

$$h_i = \gamma_i \cdot h_{i-1} + \beta_i + h_{i-1}, \quad (7)$$

- Cross-Attention layer. The ST-DiT block has been modified to include an additional Cross-Attention layer following the SSA or TSA, with the motion patches serving as the key and value to integrate with the hidden state h:

$$h_i = \text{CrossAttn}\,([h_{i-1}, f_i]) + h_{i-1}, \quad (8)$$

We evaluate three types of fusion architectures and find that the adaptive norm yields the best performance and computational efficiency. For the remainder of the paper, MGF employs the adaptive norm layer unless otherwise specified.

3.3. Data Processing and Training Strategy

Data Processing. We employ a structured data processing method to obtain high-quality training videos with consistent object motion. Initially, raw videos are segmented into shorter clips based on scene detection[2]. Subsequently, we remove invalid videos, such as those with encoding errors, zero duration, or low resolution. Furthermore, we utilize aesthetic[3] and optical flow scores [41] to filter out low-quality videos. To concentrate on the motion of primary objects, we implement camera motion filtering, using results from motion segmentation [50] and camera detection to exclude instances predominantly exhibiting camera movement. Dramatic object motions in certain videos can lead to significant optical flow deviations, which may interfere with trajectory training. To address this, we retain these videos based on a probability of $(1 - flow\_score/100)$. For eligible videos, we generate captions using the PLLaVA model [42]. During inference, we utilize GPT-4o [24] to refine prompts, ensuring alignment with training process.

Motion condition training. Inspired by DragNUWA and MotionCtrl, we adopt a two-stage training approach for trajectory learning. In the first stage, we extract dense optical flow [41] from the training video as the trajectory, providing richer information to enhance motion learning. In the second stage, we adjust the model from complete optical flow to more user-friendly trajectories by randomly selecting 1 to N object trajectories based on motion segmentation results and flow scores. To improve the scattered nature of sparse trajectories, we apply a Gaussian filter for refinement. After completing the two-stage training, Tora facilitates flexible motion control using arbitrary trajectories.

Image condition training. We employ a mask strategy to support visual conditioning. Specifically, we randomly unmask frames during training, ensuring that the video patches of unmasked frames are not subjected to any noise.

4. Experiments

4.1. Experimental Setup

Implementation Details. Tora is initialized with OpenSora v1.2 weights, and training videos have resolutions from 144p to 720p and frame counts ranging from 51 to 204. To balance training FLOP and memory usage, we adjust the batch size from 1 to 50. We use Adam Optimizer [17] with a learning rate of 2×10^{-5} on 8 NVIDIA A100. The 3D VAE is initially trained on datasets [5, 20, 21, 30] for optical flow estimation and then frozen during Tora training. We train Tora for 2 epochs with dense optical flow and fine-tune for 1 epoch with sparse trajectories. The maximum number of sampling trajectories N is set to 16. The inference step and the guidance scale are set to 30 and 7.0, respectively.

[2]https://github.com/Breakthrough/PySceneDetect

[3]https://github.com/christophschuhmann/improved-aesthetic-predictor

Dataset. Our training videos are sourced from four datasets: 1) Panda-70M [7], from which we use the training-10M subset containing high-quality videos; 2) Mixkit [10]; 3) Pexels [1]; and 4) Internal videos. The internal videos are manually annotated, with each clip labeled to include object masks and camera movement. Following our data processing pipeline, we select about 630k eligible videos for the training dataset. For inference, we curate 185 clips with diverse motion trajectories and scenes, to serve as a new benchmark for evaluating the motion controllability.

Metrics. We leverage standard metrics including Fréchet Video Distance (FVD) [34], and CLIP Similarity (CLIPSIM) [39] to quantitatively evaluate video quality. For assessing motion controllability, we utilize the Trajectory Error (TrajError) metric, which calculates the average L1 distance between the generated and predefined trajectories.

4.2. Results

We evaluate against motion-guided video generation methods using 16-frame, 64-frame, and 128-frame configurations. Trajectories are scaled proportionally for different durations. UNet-based approaches utilize sequential inference for extended generation. Since there are no DiT-based methods, we adapt DragNUWA's motion trajectory design to our foundation model as an additional baseline. Specifically, we implement its official convolutional motion feature extraction and linear projection-based injection. To ensure compatibility with base DiT model, we perform downsampling across both spatial and temporal dimensions.

As shown in Table 1, UNet-based methods all exhibit increasing deviations with longer sequences, causing motion blur and object deformation. In contrast, Tora leverages the transformer's scaling capabilities to maintain robustness, achieving 3-5 times higher trajectory accuracy and approximately 30-40% better FVD in 128-frame tests. Figure 6 shows a comparative qualitative analysis of Tora against these mainstream motion control techniques.

While the OpenSora achieves high visual quality, its inability to incorporate motion control leads to random object trajectories, resulting in significantly elevated TrajError. Notably, Tora demonstrates dual superiority over OpenSora in both motion control efficacy and visual performance at most settings. We find Tora's exceptional capacity in suppressing temporal artifacts and motion blurring compared with OpenSora (see Appendix for visual comparisons), yielding enhanced temporal coherence and video stability. While OpenSora-based DragNUWA improves motion controllability over UNet-based methods, its motion representations exhibit intrinsic incompatibility with DiT's latent space. This architectural incompatibility induces suboptimal feature integration during training, consequently degrading visual quality by about 5% below baseline.

Figure 5 presents an analysis of Trajectory Error across

Method	FVD (↓)			CLIPSIM (↑)			TrajError (↓)		
	16-frame	64-frame	128-frame	16-frame	64-frame	128-frame	16-frame	64-frame	128-frame
UNet-based method									
VideoComposer [36]	529	668	856	0.2335	0.2284	0.2236	15.11	29.14	58.76
DragNUWA [44]	475	593	784	0.2385	0.2341	0.2305	10.04	17.33	41.25
AnimateAnything [8]	487	602	775	0.2399	0.2342	0.2313	13.39	27.28	51.33
TrailBlazer [19]	459	581	756	0.2403	0.2351	0.2322	11.68	19.47	44.10
MotionCtrl [38]	463	572	731	0.2412	0.2376	0.2331	9.42	16.46	38.39
DiT-based method									
OpenSora [51]	**430**	476	533	**0.2452**	0.2433	0.2411	286.43	321.52	373.17
OpenSora-based DragNUWA*	451	504	565	0.2430	0.2419	0.2393	10.11	13.88	21.75
Tora(Ours)	438	**460**	**494**	0.2447	**0.2435**	**0.2418**	**7.23**	**8.45**	**11.72**

Table 1. Quantitative comparisons with motion-controllable video generation models. As the number of generated frames increases, Tora's performance advantage over UNet-based methods becomes more pronounced. Specifically, Tora not only enhances motion fidelity but also improves the visual quality of the foundational model. Comparisons with OpenSora-based DragNUWA highlight the strengths of our proposed motion modules, which integrate seamlessly with DiT's architecture.

Figure 5. Comparison of Trajectory Error across various resolutions and durations.

Method	FVD (↓)	CLIPSIM (↑)	TrajError (↓)
Sampling Frame	581	0.2304	27.61
Average Pooling	558	0.2325	20.97
3D VAE	**513**	**0.2358**	**14.25**

Table 2. Evaluation of the impact of different trajectory compression methods.

different resolutions and durations. Unlike UNet models, Tora shows a gradual increase in error as duration extends. This aligns with the decrease in video quality observed in the DiT model. The results demonstrate that our method effectively maintains trajectory control over longer durations.

4.3. Ablation study

Trajectory Compression. To validate the alignment of motion patches with the DiT inputs, we explore three different methods for trajectory compression, as summarized in Table 2. The first method samples the mid-frame as a keyframe for successive 4-frame intervals and uses Patch-Unshuffle [14] for spatial compression. While simple, this approach is sub-optimal for motion control due to potential flow estimation errors during rapid movements or occlusions, and the increased dissimilarity between patches complicates learning. The second method employs average pooling to gather information from successive frames. Although this captures a general sense of movement, it sacrifices precision by averaging the trajectory's direction and magnitude, diluting important motion details. Our method utilizes a 3D motion VAE to extract the global context of successive trajectory intervals. Extensive training on a large dataset of trajectory videos with this method yields the best results, highlighting the effectiveness of our customized VAE approach for compression.

Method	FVD (↓)	CLIPSIM (↑)	TrajError (↓)
Extra Channel	542	0.2329	21.07
Cross Attention	526	0.2354	18.36
Adaptive Norm	**513**	**0.2358**	**14.25**

Table 3. Different variants of motion fusion blocks employed in MGF. Adaptive Norm works best.

Block design and integrated position of MGF. We train the three variant MGF blocks as previously described, with the results presented in Table 3. Notably, the adaptive norm block achieves lowest FVD and Trajectory Error, while also exhibiting the highest computational efficiency. This advantage arises from its ability to dynamically adapt features based on varying conditions without strict alignment, a common cross-attention challenge. It also maintains temporal consistency by modulating conditional information over time, essential for incorporating motion cues. In contrast, channel concatenation can lead to information congestion, making motion signals less effective. We find that initializing the normalization layer as the identity function is vital

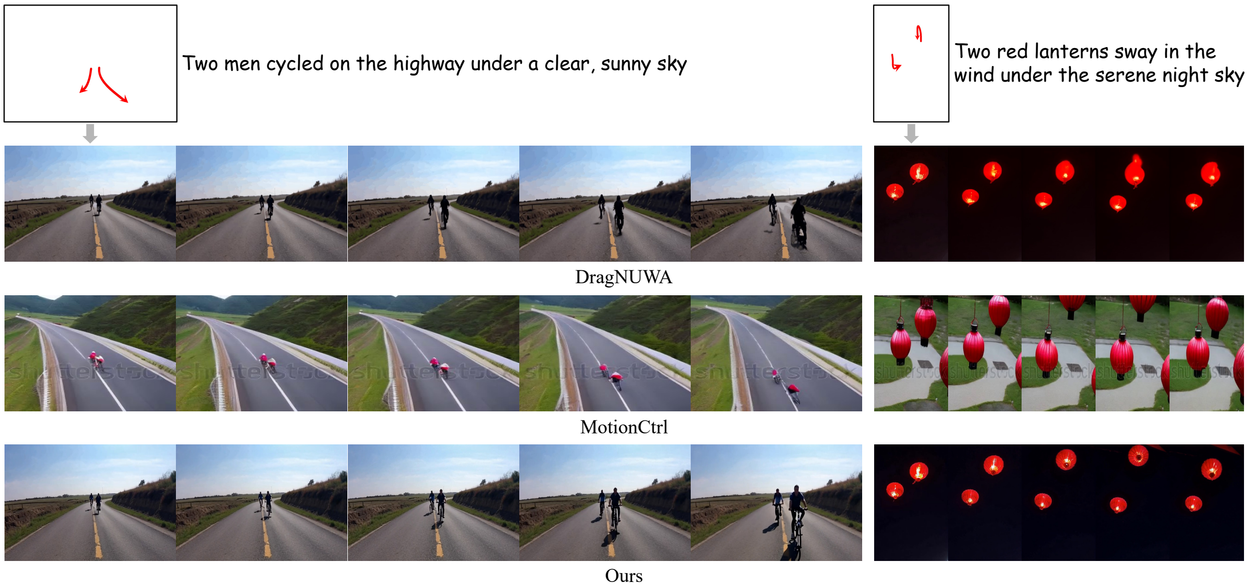

Figure 6. Qualitative Comparisons on Trajectory Control. In the bicycle scenario, Tora realistically captures pedaling motions, while other methods show legs in an unnatural, nearly horizontal position. In another case, DragNUWA causes significant deformation of the lanterns, and MotionCtrl fails to accurately depict two lanterns. Overall, Tora not only adheres precisely to the specified trajectory but also produces smoother movement that conforms to the physical world.

Motion-guidance	FVD (↓)	CLIPSIM (↑)	TrajError (↓)
Dense Flow	601	0.2307	39.34
Sparse Flow	556	0.2334	24.73
Hybrid	**513**	**0.2358**	**14.25**

Table 4. Ablation of the type of training trajectories. "Hybrid" denotes the two-stage training strategy.

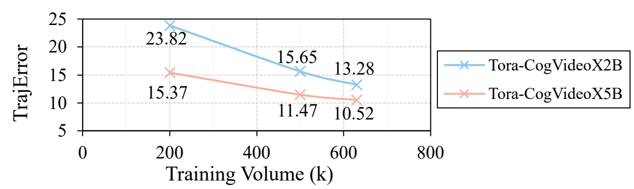

Figure 7. Scaling behavior of motion control ability in Tora.

for optimal performance. Additionally, placing the MGF module within the Temporal DiT block significantly enhances trajectory motion control, evidenced by a drop in Trajectory Error from 23.39 to 14.25.

Training Strategies. We evaluate the two-stage training approach, with results in Table 4. Training only with dense optical flow is ineffective, as it fails to capture the details of sparse trajectories. Conversely, using only sparse trajectories provides limited information, complicating the learning process. In contrast, our two-stage strategy demonstrates better adaptability and versatility in managing various motion patterns, leading to improved overall performance.

Scaling motion-control ability. We investigate scaling laws in motion-controllable video generation by examining model parameter size and training volume. To leverage existing multi-size foundational models, we transfer our motion modules to CogVideoX architectures (2B/5B parameters), resulting in Tora-CogVideoX2B (2.5B) and Tora-CogVideoX5B (6.3B). Our implementation retains base VAE for motion compression while inserting MGF prior to Expert Transformer's full attention module. Figure 7 demonstrates that increasing model scale and training data improves motion control well, confirming our modules' seamless compatibility with DiT's scalability framework.

5. Conclusion

This paper introduces Tora, the first trajectory-oriented Diffusion Transformer framework for video generation. Tora effectively encodes arbitrary trajectories into spacetime motion patches, which align well with the scaling properties of DiT, thereby enabling more realistic simulations of physical world movements. By employing a two-stage training process, Tora achieves motion-controllable video generation across a wide range of durations, aspect ratios, and resolutions. Remarkably, it can generate high-quality videos that adhere to specified trajectories, producing up to 204 frames at 720p resolution. This capability underscores Tora's versatility and robustness in handling diverse motion patterns while maintaining high visual fidelity. We hope our work provides a strong baseline for future research in motion-guided Diffusion Transformer methods.

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